

A Direct-to-Chip Liquid-Cooled Twin for AI Data Halls

Coupling that follows the plumbing, not the floor plan

The air-cooled twin put a ceiling on what this paper could claim: neighbouring racks' inlet temperatures correlate at 0.9953, so spatial context adds almost nothing to single-rack accuracy and a graph model has little room to win. This report describes a liquid-cooled hall built to test whether that ceiling is a property of data halls or a property of air.

1. What is and is not real

No part of this thermal model comes from a real data centre. The only measured data anywhere in this project is the Alibaba PAI GPU workload trace. Every thermal number below is ours.

More sharply: the headline result — that racks on different coolant loops show *exactly zero* coupling — is not a discovery. It is a consequence of how the model is written. The CDU approach temperature is a function of that CDU's own duty, so racks on different loops cannot influence each other, and the probe then measures that they do not. A reviewer will see this immediately, and the paper must not present it as an empirical finding.

The air twin was less circular in this respect. There, $D = K^{-1}[(I - A^T)^{-1} - I]$ produced multi-hop coupling that was never specified: the 70 / 21 / 6 % hop profile emerged from a matrix inverse over a local kernel. The liquid model has much less of that. The one genuinely emergent part is the ordering within a branch produced by manifold cross-talk, which is why loop topology explains 0.645 of the coupling variance rather than 1.000. That residual is the only part of the structure doing real work.

Tier	What	Status
Measured	Alibaba PAI GPU workload trace, 100,000 jobs over 289.8 h	real
Structural	CDU / branch manifold / cold plate architecture; the roughly 80:20 liquid-air heat split; ASHRAE W32–W45 liquid classes; ~77 kW racks and an 85 °C case limit, the right order for GB200-class hardware	real, uncited
Plausible	Design ΔT 12 K; 90 L/min per rack; CDU approach 2–8 K. Inside published vendor envelopes, but not read off a datasheet and not fitted to anything	unvalidated
Invented	manifold_crosstalk 0.16; flow_droop_per_m 0.006; cdu_approach_gain 3.0; case_flow_exponent 0.40. Chosen to produce behaviour of the right shape	no source

In reality racks on different CDUs *are* weakly coupled, through the shared facility water loop, the shared air volume and shared pumping plant. The air path is modelled; the facility-water return is not, so the reported separation is cleaner than a real hall's would be.

2. Why liquid cooling changes the question

Air recirculation is diffuse. Every rack in a region shares one thermal environment, so a rack's own temperature already encodes what its neighbours are experiencing, and neighbour features are close to redundant. That is the 0.9953 correlation, and it is why the edge-shuffle ablation on the air twin was expected to come back near-null.

A direct-to-chip hall is plumbed, not ventilated. Coolant leaves a Coolant Distribution Unit, runs along a branch manifold, and is drawn off in parallel by the racks on that branch. Racks sharing a branch share a supply temperature, a flow budget and a CDU. Racks on a different loop share almost nothing — however close they stand.

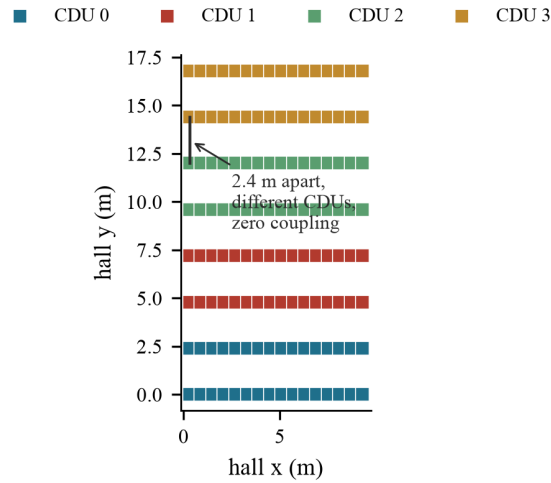


Figure 1. hall_11: 128 racks, four CDUs, two branch manifolds each. Colour is coolant loop, not position. Rows that sit next to each other on the floor belong to different loops, and the marked pair stands 2.4 m apart with no hydraulic connection at all.

3. The model

Cold plates take 80% of rack power into coolant; the remaining fifth leaves as air and recirculates through the air twin's kernel unchanged, with rack airflow scaled to the air share because a liquid-cooled rack ships far smaller fans. At each CDU the secondary supply sits above the facility water by an approach that degrades as the unit approaches its rated duty, which is what couples every rack on a loop to every other rack on it. Along a branch, flow droops with distance from the feed and the supply is warmed by the return of every rack upstream.

Two errors found by measuring rather than assuming

Cross-talk is what makes the graph non-trivial. The first version had only a static manifold gain, so perturbing any rack moved every rack on its loop by exactly the same amount, and loop topology explained **100%** of the coupling variance. That is not a graph problem: one categorical “which CDU” feature would capture the whole structure. Adding supply-return manifold cross-talk made the coupling directed and ordered within a branch, and explained variance fell to 0.645.

The case-to-coolant rise is per device, not per rack. It was first written as kelvin per kilowatt of *rack* power, which made a denser rack appear to run each accelerator hotter. Every accelerator has its own cold plate fed in parallel off the rack manifold, so the rise is set by per-device power.

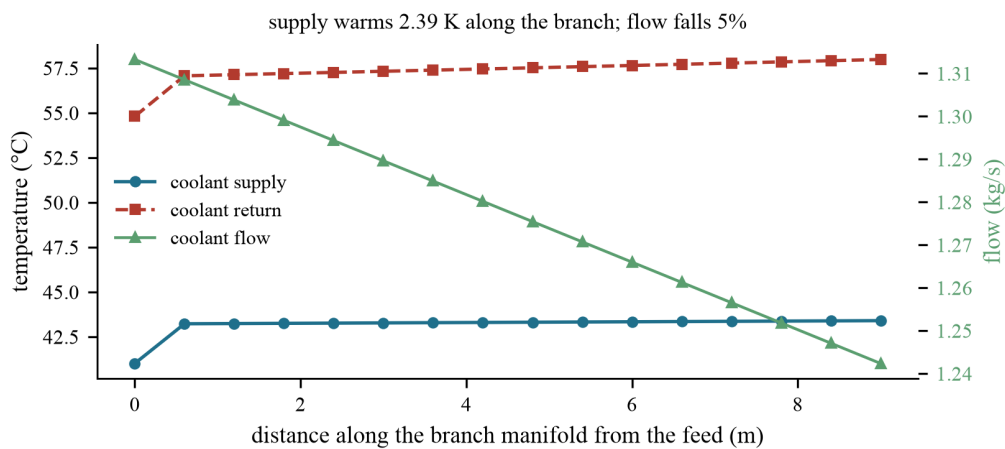


Figure 2. Along one branch manifold at 90% utilisation, 34 °C facility water, pumps at 85%. Supply warms with distance from the feed because the return manifold alongside it carries the heat of every upstream rack; flow falls over the same span. A rack at the far end is therefore penalised twice.

4. What the probe measures

Each rack is given a 5 kW step and the coolant supply response is measured at every other rack. hall_11 at 85% utilisation, 34 °C facility water, pumps at 85%.

Relationship	Pairs	Mean $ \Delta T $ (K)	Mean gap (m)
same branch (shared manifold)	1,920	0.01817	3.40
same CDU, different branch	2,048	0.00960	4.24
different CDU	12,288	0.00000	8.99

The most strongly coupled group is not the physically closest. Restricting to racks within one row pitch of each other makes the point sharper:

Physically within 2.6 m	Pairs	Mean $ \Delta T $ (K)
on the same CDU	1,232	0.01753
on different CDUs	276	0.00000

276 pairs of racks stand side by side and are thermally independent. A $k=4$ neighbour feature set built on Euclidean distance would select those racks and learn from channels carrying no signal. Over all pairs, coupling correlates with distance at -0.448 , while loop topology explains 0.645 of its variance.

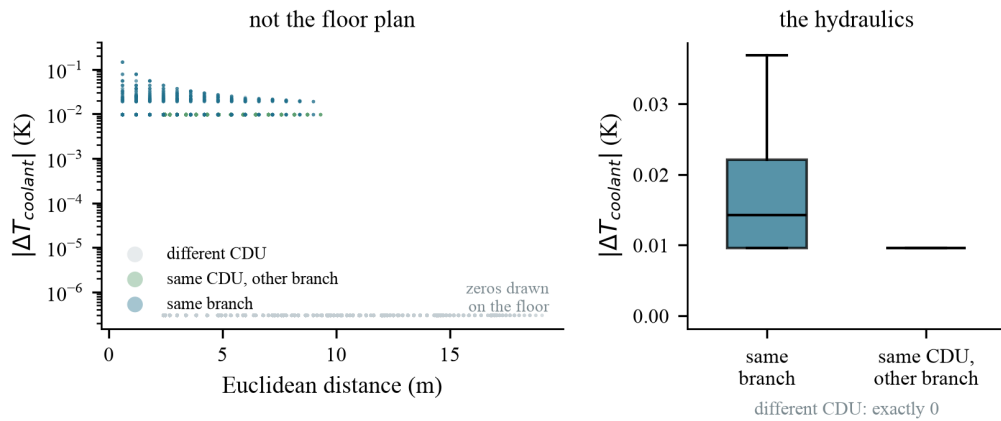


Figure 3. Left: coupling against physical distance, coloured by hydraulic relationship. The three bands are flat — distance carries almost no information about coupling. Zeros are drawn on the axis floor because the scale is logarithmic. Right: the same values grouped by relationship.

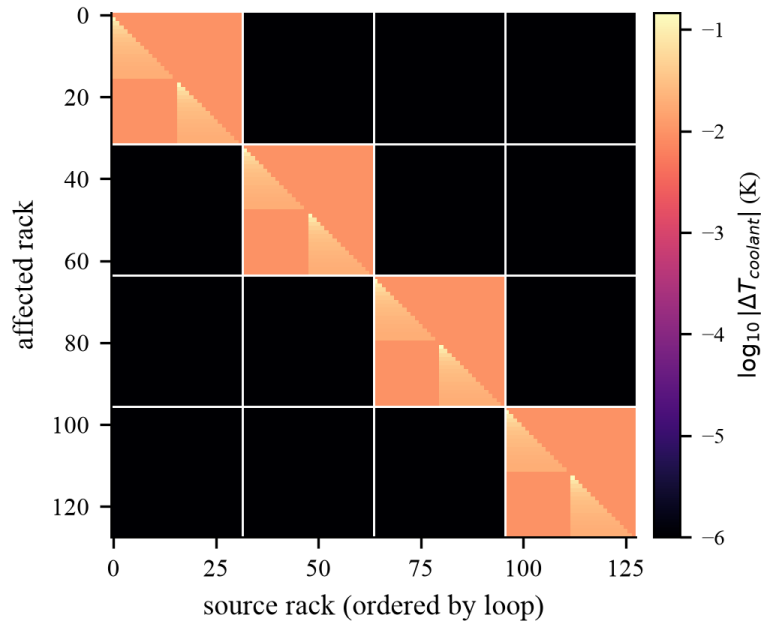


Figure 4. The influence matrix with racks ordered by loop and then by manifold position. Four blocks, nothing between them. The triangular shading inside each block is the cross-talk term: upstream racks warm downstream ones and not the reverse. This figure also makes the circularity of Section 1 plain — the black off-diagonal blocks are imposed, not measured.

5. Operating envelope

Across the ASHRAE liquid classes W32 to W45. At the cold end the hall has easy margin; at the warm end, where a hall with no mechanical chilling actually runs, the 85 °C case limit binds and pump speed is what saves it. This gives the cooling controller two levers with different costs, which the air hall did not have: facility water temperature is slow and plant-wide, while pump speed is fast, per-loop, and moves chip temperature through the cold plate's convective coefficient without moving the coolant supply at the head of a branch at all.

Load	Facility water	Pumps	Coolant supply	Peak case	Throttling
90%	26 °C	100%	33.2	60.4	no
90%	34 °C	100%	41.3	69.5	no
90%	42 °C	100%	49.5	78.6	no
100%	42 °C	80%	50.0	84.6	no
100%	42 °C	55%	50.1	91.1	yes
60%	42 °C	55%	48.4	77.6	no

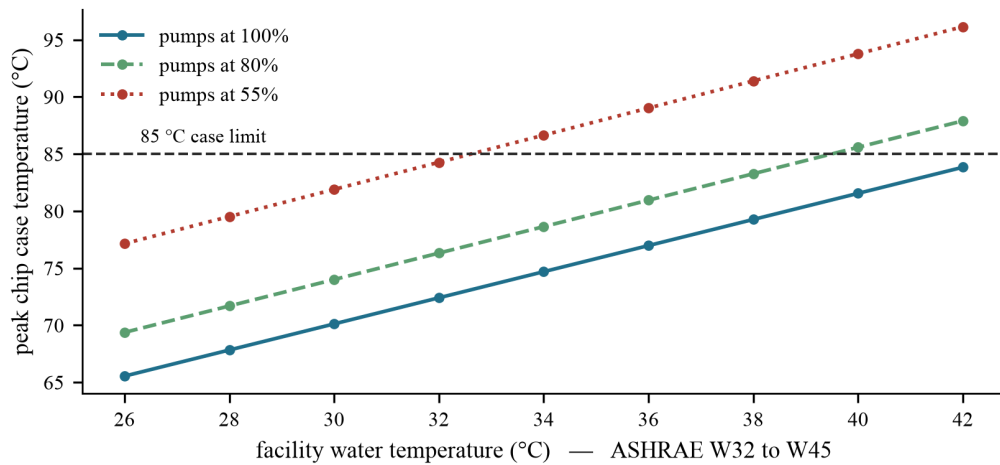


Figure 5. Peak chip case temperature at full load across the warm-water band, by pump speed. The three curves are nearly parallel, which is the point: pump speed shifts the whole envelope, so a hall can trade pumping energy against how warm its facility water is allowed to run.

6. What this changes for the paper

#	Consequence
1	The accuracy experiment has room again. The air hall's 0.995 neighbour correlation capped what any spatial model could add. Here the most strongly coupled racks are ones a distance-based model would not select at all.
2	The edge-shuffle ablation becomes a real test. Shuffling edges in the air hall was expected to cost almost nothing. Here it should destroy the model, because the edges carry the plumbing and nothing else encodes it.
3	The directed-edge ablation is no longer cosmetic. Manifold cross-talk runs strictly upstream to downstream, so reversing an edge is physically wrong rather than merely uninformative.
4	Placement gains a sharper lever. Where to put a job becomes partly a question of which loop has headroom, which is invisible to any floor-plan heuristic.

7. Before this becomes a claim

Four things should happen before any of Section 4 appears in a paper. The first two address the circularity in Section 1 directly.

#	Action	Why
1	Model the shared facility water loop	Cross-CDU coupling becomes small but non-zero rather than stipulated zero. Cheapest fix, and it removes the worst of the circularity.
2	Sweep manifold_crosstalk	If the graph model's advantage exists at 0.16 and vanishes at 0.05, the result is an artefact of a number with no source.
3	Anchor the CDU curve to a datasheet	Vendors publish approach-against-load curves. This converts an invented parameter into a cited one.
4	State the framing precisely	Not “evaluated on a liquid-cooled data hall” but “evaluated on a parametric direct-to-chip model whose coupling structure is imposed by construction”.

Known limitations of the model as it stands

Racks hang in parallel off a manifold rather than in a series cascade; immersion tanks and some rear-door designs are genuinely serial and would show stronger ordering still. Flow is set by manifold position rather than solved hydraulically, so a rack drawing more does not starve its neighbours. Single-phase only, with no two-phase or evaporative cold plates. The CDU approach is a smooth ramp in duty rather than a heat-exchanger effectiveness curve with a real NTU.